# NREPESH JOSHI

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Machine Learning Engineer with expertise in developing and deploying high-impact ML solutions in healthcare and legal industries. Proven track record in leading teams, optimizing ML workflows, and delivering revenue-driving products. Experienced in LLMs, NLP, and Computer Vision. Passionate about building efficient, scalable ML systems that drive business growth and innovation.

## PROFESSIONAL EXPERIENCE

### TACKLE AI, Schaumburg IL (Oct 2020 - Present)

TackleAI, a $35M seed-funded leader revolutionizing document intelligence for healthcare and legal industries. I now lead the ML team, report directly to the CTO, and drive hiring, optimization, and revenue-generating product launches.

### Machine Learning Team Lead (Mar 2023 - Present)

- Led a team of 5 ML engineers, driving 90% of company revenue in a critical quarter and sustaining 60% of total revenue through high-value solutions for top-tier legal clients.
- Processed over 3M+ medical records containing PHI, ensuring HIPAA-compliant workflows and leveraging multimodal classification (Layoutlmv3) and vector similarity search to set new standards in high-volume, on-premise data processing, and integrated new chatbot functionality.
- Created 3 essential packages—OCR XML integration, multithreaded file management, and agentic LLM templates—reducing developer time by 70% and streamlining workflows across the team.
- Enhanced generative AI LLM workflows with custom similarity search for a 25x speed boost by removing external dependencies (e.g., LangChain, vector DBs), sustaining 90% GPU utilization for 30+ hours with minimal errors.
- Built asynchronous backend architecture leveraging gRPC endpoints and WebSockets, orchestrating real-time inference via Redis, Celery, and multimodal LLMs on vLLM—delivering enhanced OCR JSON responses for a mission-critical company initiative.
- Engineered a proprietary OCR solution with 60% automation and 94% accuracy, simplifying data extraction from complex document formats using multimodal LLMs and Vision Transformers (ViT).
- Leveraged synthetic data augmentation to improve object detection model performance by 30%, enabling accurate redaction of sensitive information in breached documents for legal casework.

### Founding Machine Learning Engineer (Oct 2020 - Mar 2023)

- Spearheaded development of 85% of core computer vision models (bounding box, instance, and semantic segmentation), critical to securing seed+ funding and addressing high-value client requirements.
- Created an optimized image similarity search model leveraging feature-extractor embeddings and custom LSH vector spaces, achieving 92% accuracy and an 84.5% speed increase across 1M+ images.
- Fine-tuned Spacy DistilBERT for Named Entity Recognition (NER), achieving (+5% precision, +8% recall, F1: 97%), significantly improving extraction accuracy.
- Migrated Spacy NLP models to Triton Server, reducing vRAM usage by 10x, enhancing hosting efficiency, and freeing resources for further model deployments.

- Reduced manual labeling workload by 85% through advanced unsupervised clustering (HDBSCAN), dimensionality reduction, and data standardization.
- Established CI/CD pipelines for deploying ML models via gRPC and REST APIs on AWS, utilizing ECS, ECR, and Lambda for scalable inference solutions.

## EDUCATION AND CERTIFICATION

**GEORGIA INSTITUTE OF TECHNOLOGY, Atlanta, GA** January 2026 – Present
Master of Science in Computer Science; Machine Learning Specialization

**KNOX COLLEGE, Galesburg, IL**
Bachelor of Science; Double Major in Computer Science & Physics
Magna Cum Laude ~ Cumulative GPA 3.77/4.0 ~ CS Major GPA 4.0/4.0
Scholarships; Ellen Browning Scripps, Service & Leadership and Knox Grant

## COMMUNITY INVOLVEMENT

- **Speaker, AI Conference for a Prosperous Nepal –** Presented "AI & ML in Government Projects," covering practical AI applications for societal benefit and addressing data and computational challenges.
- **ODSC Boston 2024 Attendee –** Engaged in AI workshops and discussions with industry leaders.
- **AI Tinkerers Chicago Meetups –** Active participant in AI community meetups exploring trends and applications.
- **Chicago In-Context MCP Mini-Hackathon 2025 – 2nd Place Winner -** Developed FastMCP server enabling LLM tool calling to query Chicago building heights and recommend offline map downloads based on urban density analysis.

## SKILLS

- **Programming Languages:** Python, Java, C#
- **Machine Learning Frameworks:** Tensorflow, Keras, Pytorch, HuggingFace, Spacy, Langchain, LlamaIndex, OpenAI
- **Dimensionality Reduction:** PCA, TSNE, UMAP
- **Web Frameworks and Databases:** Flask, FastApi, Nginx, S3, MySQL, MongoDB, Redis, Milvus
- **Platforms:** Amazon Web Services (AWS), Google Cloud Platform (GCP), Linux
- **Containers:** Docker/Docker compose, Kubernetes

## TECHNICAL PROJECTS

### Plot Point AI-Powered Active Reading Assistant

- Built an AI-driven active reading assistant that enables real-time question answering and contextual insights within PDFs, eliminating the need for external tools.
- Implemented co-reference resolution using OpenAI's gpt-4-o-mini with advanced prompt engineering to ensure natural language understanding of pronouns and references within documents.
- Developed a high-performance similarity search engine using Annoy (Approximate Nearest Neighbors) for on-the-fly document querying, ensuring fast and accurate text retrieval.
- Designed and optimized the backend in Python, handling API interactions, document parsing, and conversation flow seamlessly hosting the product in Render.
- Integrated OpenAI’s LLM for contextual Q&A, enabling users to ask questions naturally while reading, with the AI understanding references like “he,” “she,” or “it” in context.